

INNOVATION PROJECT AND INTERNSHIP PRESENTATION

BY EMILIE JIANG
SUPERVISED BY XIAO XIAO

SEPTEMBER 2025

CREATIVE TECHNOLOGY
2025-2026

“ Use artificial intelligence to help nuclear fusion become a controllable, a clean and unlimited energy source for the future “



浙江大学 杭州国际科创中心
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DIMENET++ TRANSFORMER PROJECT

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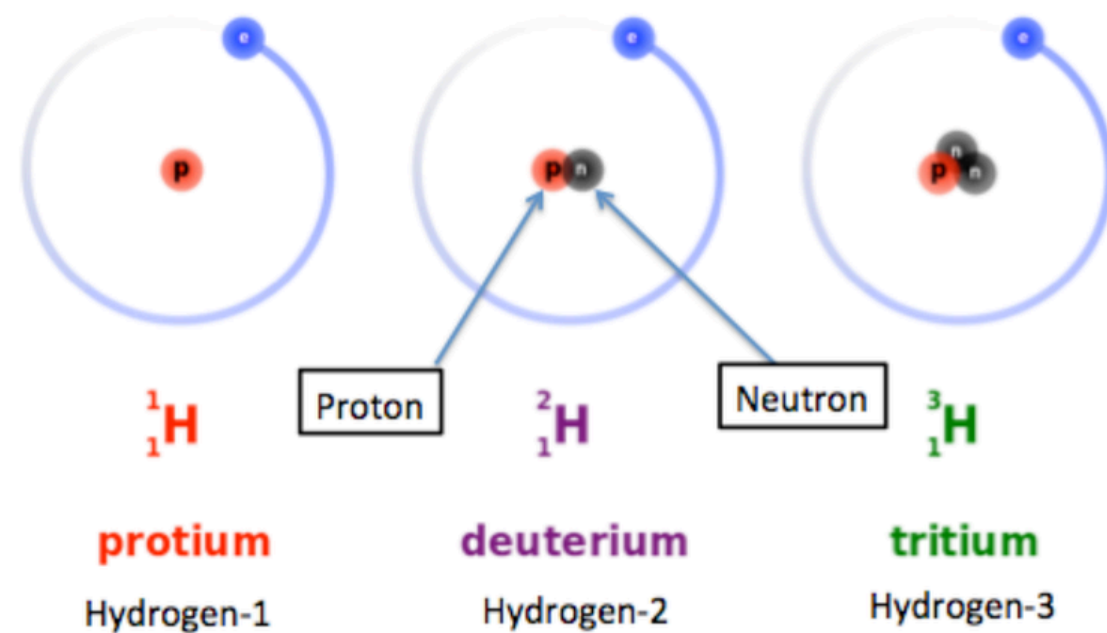
APRIL - AUGUST
2025

THE INTERNATIONAL THERMONUCLEAR EXPERIMENTAL REACTOR (ITER)



<https://www.nytimes.com/2017/03/27/science/fusion-power-plant-iter-france.html>

THREE ISOTOPES OF HYDROGEN

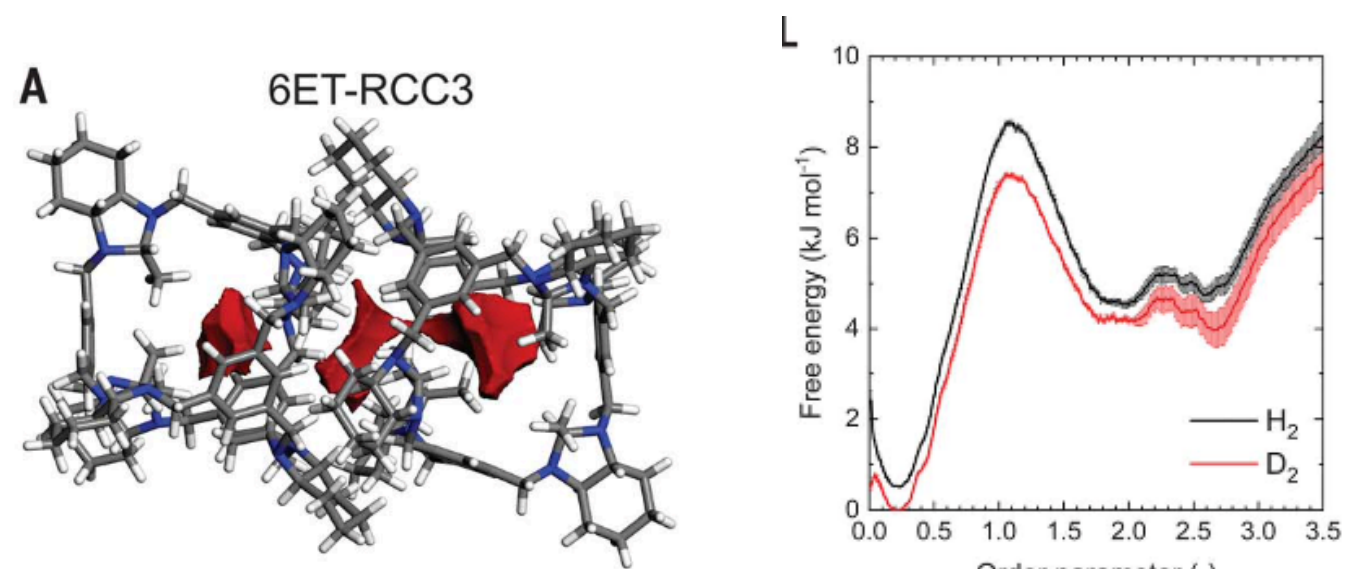


Isotopes of Hydrogen	Number of Protons	Number of Neutrons	Atomic Mass	Symbol
Protium <i>or</i> Hydrogen-1	1	0	1.00782504 or 1	${}^1_1\text{H}$ or ${}^1\text{H}$
Deuterium <i>or</i> Hydrogen-2	1	1	2.01410178 or 2	${}^2_1\text{H}$ or ${}^2\text{H}$
Tritium <i>or</i> Hydrogen-3	1	2	3.0160492 or 3	${}^3_1\text{H}$ or ${}^3\text{H}$

Property	Hydrogen	Deuterium	Tritium
Relative abundance (%)	99.985	0.0156	10^{-15}

BARELY POROUS ORGANIC CAGES FOR HYDROGEN ISOTOPE SEPARATION (2019)

BY MING LIU, LINDA ZHANG, MARC A LITTLE, VENKAT KAPIL, MICHELE CERIOTTI, SIYUAN YANG, ...



RESEARCH

ISOTOPIC SEPARATION

Barely porous organic cages for hydrogen isotope separation

Ming Liu¹, Linda Zhang², Marc A. Little¹, Venkat Kapil³, Michele Ceriotti³, Siyuan Yang⁴, Lifeng Ding⁴, Daniel L. Holden¹, Rafael Balderas-Xicohtencatl², Donglin He¹, Rob Clowes¹, Samantha Y. Chong¹, Gisela Schütz², Linjiang Chen^{1,5}, Michael Hirscher^{2,*}, Andrew I. Cooper^{1,5*}

The separation of hydrogen isotopes for applications such as nuclear fusion is a major challenge. Current technologies are energy intensive and inefficient. Nanoporous materials have the potential to separate hydrogen isotopes by kinetic quantum sieving, but high separation selectivity tends to correlate with low adsorption capacity, which can prohibit process scale-up. In this study, we use organic synthesis to modify the internal cavities of cage molecules to produce hybrid materials that are excellent quantum sieves. By combining small-pore and large-pore cages together in a single solid, we produce a material with optimal separation performance that combines an excellent deuterium/hydrogen selectivity (8.0) with a high deuterium uptake (4.7 millimoles per gram).

Deuterium (D) is used as a neutron moderator, as a nonradioactive isotopic tracer, and in neutron scattering experiments. These applications need high-purity deuterium, which is expensive because of its low natural abundance (0.0156 mol %). Typically, D₂ is produced by electrolysis of heavy water and extracted through the Girdler-sulfide process (1) or by cryogenic distillation at 24 K (2). Both processes are costly and energy intensive because of the multiple enrichment steps required (3).

An attractive alternative to purify D₂ from H₂/D₂ gas mixtures is to adsorb D₂ selectively on a microporous bed (4). Kinetic quantum sieving (KQS) with nanoporous solids was first proposed by Beenakker *et al.* (5). The KQS effect becomes pronounced when the difference in size between a hydrogen molecule and the confining space is comparable to the thermal de Broglie wavelength of molecular hydrogen, $\lambda_T = h/(2\pi mk_B T)^{1/2}$ (h , Planck's constant; m , mass; k_B , Boltzmann constant; T , temperature). Quantum sieving has been exploited for the separation of gaseous isotope mixtures such as D₂/H₂ (6), but it is challenging to identify suitable porous solids. This is because KQS requires ultrafine pore apertures (~3 Å) (7, 8), and this typically leads to materials with low pore volumes and, hence, low D₂ adsorption capacities, making such processes difficult to scale. An

analogous selectivity-capacity trade-off—or, for membranes, a selectivity-permeance trade-off—is observed for a wide range of other gas separations that do not involve KQS (9–12).

Various porous materials have been studied for hydrogen isotope separation, including carbons (13, 14), carbon nanotubes (15), zeolites (16, 17), metal-organic frameworks (MOFs) (6, 18, 19), covalent organic frameworks (COFs) (20), and 2D crystals (21). MOFs and COFs have received much attention because of their crystalline nature and synthetically tunable pore size and functionality (22). However, even with MOFs or COFs, it is challenging to tune pore sizes at the exceptionally fine level required for KQS. For example, in MOFs, a common strategy for tuning the pore apertures is through systematic expansion or reduction of the number of the phenylene rings in the organic linker, with an increment or a decrement of ~2.8 Å (23); that is, too coarse for fine-tuning the aperture for optimum KQS.

Porous organic cages (POCs) (24, 25) are discrete molecules that have been used previously for the separation of xylene isomers (26), noble gases (27), and chiral molecules (27). POCs might also be promising candidates for H₂/D₂ separation but, unlike MOFs and COFs, it is difficult to tune the pore size in POCs simply by changing the constituent linkers. Because POCs are discrete molecules, small changes in their molecular structure can have profound effects on the solid-state packing of the cages and, hence, the pore structure (24). Modifying only the interior of the cage molecule, rather than the cage structure itself, could allow us to avoid altering the crystal packing. Internal functionalization is possible. For example, Mastalerz and co-workers described the postsynthetic modification of the interior of organic cages by using a sixfold Williamson etherification (28), but this interior modification affected the cage shape, which in turn altered the crystal packing mode.

Systematically tuning the pore size of organic cages by postsynthetic modification

We synthesized a series of internally post-functionalized porous organic cages that crystallized in an isostructural way. Specifically, we used a protection-deprotection strategy to produce cages where five out of the six internal reaction sites in the cage cavity were functionalized by means of a formaldehyde “tying” method we reported previously (29) (Fig. 1, A and B). After deprotection, the single unreacted diamine group can be reacted with a range of aldehyde or ketone precursors to achieve ultra-fine control over the cage cavity size and, hence, the pore envelope (Fig. 1C). This method allowed us to tune the size and shape of the cage pore systematically at the atomic level without affecting the external cage shape and size or its crystal packing. Using this strategy, we tuned the pore size for the series of POCs in the range 1.95 to 3.5 Å, which is a useful cut-off range for the separation of gas pairs such as H₂/N₂, H₂/CO, CH₄/N₂, and Xe/Kr (7).

Previously, we reported that an organic cage with six diamine groups, **RCC3**, reacts with six formaldehyde molecules to form **6FT-RCC3** through the formation of cyclic aminal rings (29). **RCC3** was reacted with six equivalents of acetaldehyde to form a new cage, **6ET-RCC3** (reaction *a*, Fig. 1B), where bulkier ethylidene bridges replaced the six methylene bridges in **6FT-RCC3**. By contrast, **RCC3** reacted with just a single acetone molecule to form **1AT-RCC3** (29) (reaction *b*, Fig. 1B); steric hindrance prevented further reaction in the cage cavity.

On the basis of this result, we developed a protection-deprotection strategy in which we reacted the five unfunctionalized diamine groups in **1AT-RCC3** with gaseous formaldehyde in the solid state, through a single-crystal-to-single-crystal reaction, to form a new dual “tied” cage molecule, **1AT-5FT-RCC3** (reaction *d*, Fig. 1B and fig. S1). Solid-state synthesis was essential, in that when **1AT-RCC3** was mixed with formaldehyde in solution (CHCl₃ or MeOH), the single acetone protecting group was displaced to afford **6FT-RCC3** (reaction *c* in Fig. 1B and fig. S2). By exploiting the different chemical liabilities of imidazolidine rings in **1AT-5FT-RCC3**, we could selectively hydrolyze the single propyl “tie” by stirring in CHCl₃:MeOH (1:1, v/v), thus synthesizing a new deprotected cage, **5FT-RCC3** (reaction *e* in Fig. 1B and fig. S3). After deprotection, we reacted the now vacant diamine group in **5FT-RCC3** with acetaldehyde and propionaldehyde to synthesize two new cages, **1ET-5FT-RCC3** and **1PT-5FT-RCC3** (reaction *f*, Fig. 1B). All reactions proceeded with nearly 100% conversion with no additional purification steps.

The parent cage for this study, **RCC3**, is derived from the shape-persistent imine precursor, **CC3**, which crystallized from most organic solvents to form a microporous solid

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MCRT

self-supervised pre-training (Masked Atom Prediction - MAP)

→ prediction of specific molecular properties

GOAL :

train molecular graphs to reconstruct their hidden atomic types

use generated embeddings for downstream tasks
(useful for chemistry or materials)

Chemical
Science

EDGE ARTICLE

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A universal foundation model for transfer learning in molecular crystals†

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The physical and chemical properties of molecular crystals are a combined function of molecular structure and the molecular crystal packing. Specific crystal packings can enable applications such as pharmaceuticals, organic electronics, and porous materials for gas storage. However, to design such materials, we need to predict both crystal structure and the resulting physical properties, and this is expensive using traditional computational methods. Machine-learned interatomic potential methods offer major accelerations here, but molecular crystal structure prediction remains challenging due to the weak intermolecular interactions that dictate crystal packing. Moreover, machine-learned interatomic potentials do not accelerate the prediction of all physical properties for molecular crystals. Here we present Molecular Crystal Representation from Transformers (MCRT), a transformer-based model for molecular crystal property prediction that is pre-trained on 706 126 experimental crystal structures extracted from the Cambridge Structural Database (CSD). MCRT employs four different pre-training tasks to extract both local and global representations from the crystals using multi-modal features to encode crystal structure and geometry. MCRT has the potential to serve as a universal foundation model for predicting a range of properties for molecular crystals, achieving state-of-the-art results even when fine-tuned on small-scale datasets. We demonstrate MCRT's practical utility in both crystal property prediction and crystal structure prediction. We also show that model predictions can be interpreted by using attention scores.

1 Introduction

Molecular crystals have diverse applications including pharmaceuticals,¹ organic electronics,² optical materials,³ and materials for gas storage and separation.^{4–6} In all cases, the properties of molecular crystals depend on the crystal packing. For example, pharmaceutical molecules can have widely different solubilities depending on the crystalline form, and in organic electronics, charge transport is critically dependent on crystal packing. However, molecular crystal packing is notoriously difficult to predict because it is dictated by a range of weak intermolecular interactions, such as van der Waals forces, aromatic pi-stacking, and hydrogen bonds.⁷ This is a major hurdle for digital material design because if we cannot predict crystal structure then we cannot, by definition, predict the functional properties of the crystal. To address this challenge,

crystal structure prediction (CSP) methods have been created to identify molecular crystals with specific target functionalities. For example, energy–structure–function (ESF) maps have guided the synthesis of various functional molecular crystals.^{6–10} However, despite these successes, calculating the physical properties for each structure on an ESF map, or even a sub-set of low-energy structures, can be computationally demanding. This problem is two-fold: the prediction of lattice energy, or crystal stability, is itself computationally expensive, and the functional property calculations are usually even more expensive. To tackle this, there has been a surge of interest in machine learning (ML) techniques for the rapid prediction of materials properties and the elucidation of structure-property relationships^{11,12} at a fraction of the cost of first-principles methods, such as density functional theory (DFT).

Learning accurate representations is a crucial aspect of machine learning theory that also extends to learning molecular representations. Different types of materials pose different challenges when learning accurate latent representations. For example, in solid-state systems it is essential to capture features such as long-range interactions and periodicity in property prediction tasks.¹³ This is especially challenging in organic molecular crystals due to the intermolecular interactions,⁷ which are typically weaker than for ionic inorganic materials.

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† Electronic supplementary information (ESI) available. See DOI: <https://doi.org/10.1039/d5sc00677e>

‡ These authors contributed equally to this work.

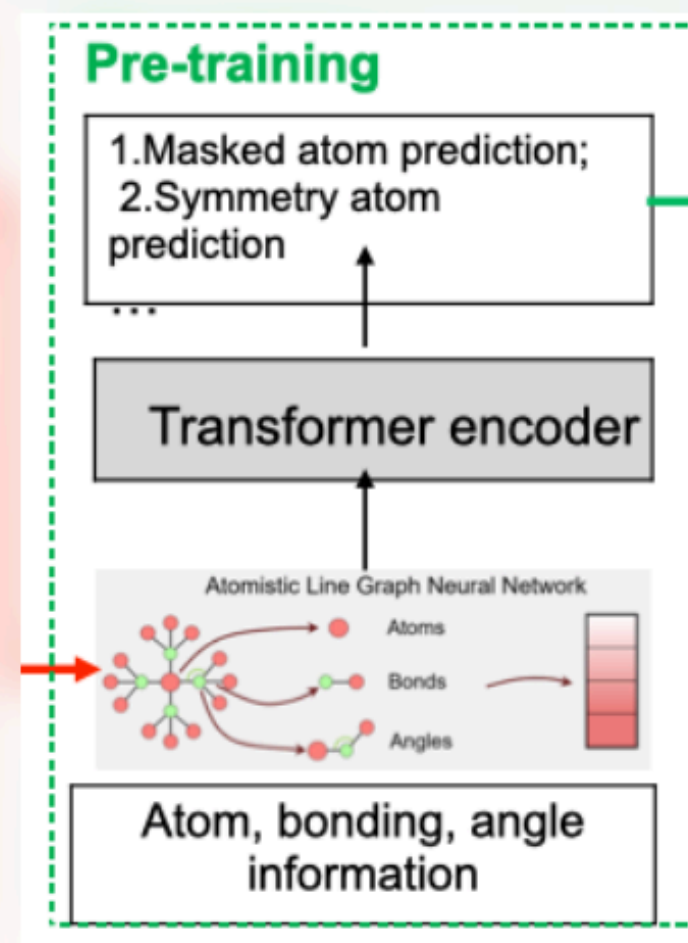
12844 | *Chem. Sci.*, 2025, 16, 12844–12859

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<https://github.com/fmgggggg/MCRT>

Feng, M., Zhao, C., Day, G. M., Evangelopoulos, X. & Cooper, A. I. (2025). A universal foundation model for transfer learning in molecular crystals. *Chemical Science*.



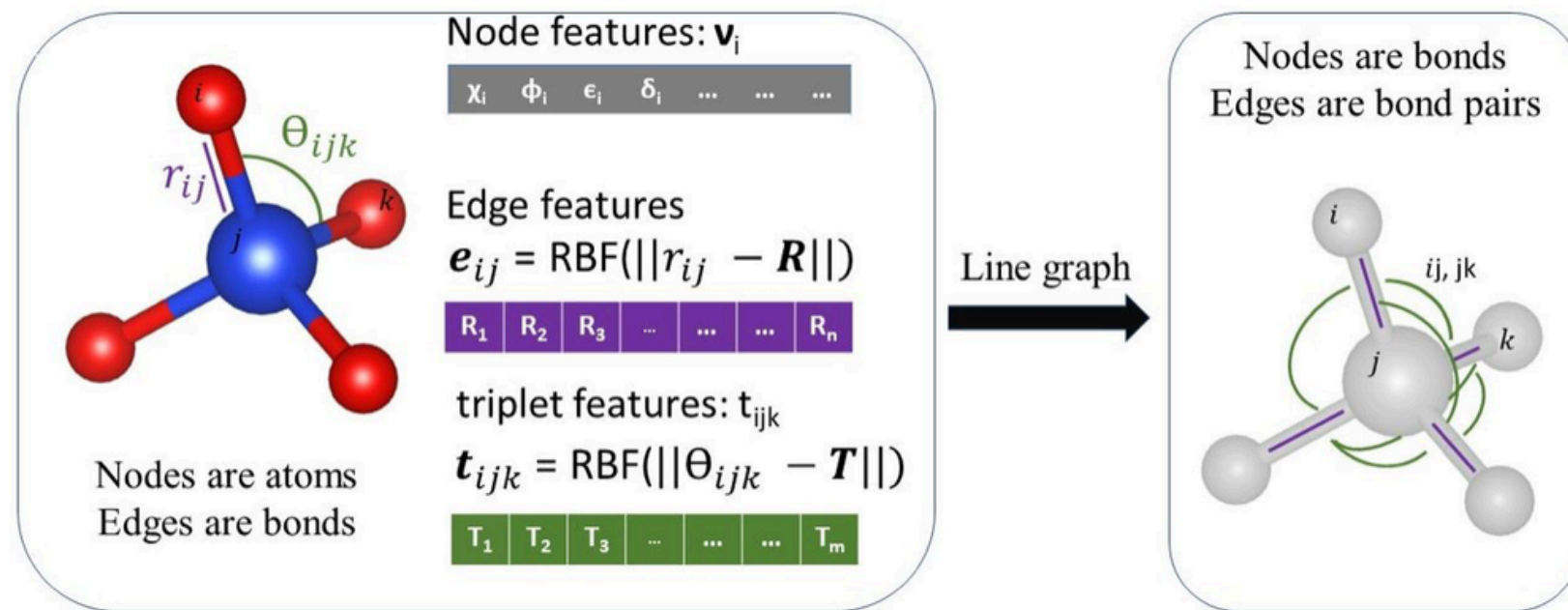
ALIGNN

(Atomistic Line Graph Neural Network)

→ PREDICT MATERIAL PROPERTIES

DUAL GRAPH REPRESENTATION :

- BOND GRAPH :
 - ATOMS AS NODE
 - BONDS AS EDGES
- LINE GRAPH :
 - NODES = BONDS FROM BOND GRAPH
 - EDGES = BOND ANGLES



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ARTICLE OPEN

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Atomistic Line Graph Neural Network for improved materials property predictions

Kamal Choudhary^{1,2,3,4} and Brian DeCost^{1,4}

Graph neural networks (GNN) have been shown to provide substantial performance improvements for atomistic material representation and modeling compared with descriptor-based machine learning models. While most existing GNN models for atomistic predictions are based on atomic distance information, they do not explicitly incorporate bond angles, which are critical for distinguishing many atomic structures. Furthermore, many material properties are known to be sensitive to slight changes in bond angles. We present an Atomistic Line Graph Neural Network (ALIGNN), a GNN architecture that performs message passing on both the interatomic bond graph and its line graph corresponding to bond angles. We demonstrate that angle information can be explicitly and efficiently included, leading to improved performance on multiple atomistic prediction tasks. We ALIGNN models for predicting 52 solid-state and molecular properties available in the JARVIS-DFT, Materials project, and QM9 databases. ALIGNN can outperform some previously reported GNN models on atomistic prediction tasks with better or comparable model training speed.

npj Computational Materials (2021)7:185; <https://doi.org/10.1038/s41524-021-00650-1>

INTRODUCTION

Graphs are a powerful non-Euclidean data structure method for establishing relationships between features (nodes) and their relationships (edges)^{1,2}. Graph neural networks (GNN)^{3,4} have immense potential for modeling complex phenomena. Common applications of GNNs include community detection and link prediction in social networks^{5,6}, functional time series on brain structures⁷, gene DNA on regulatory networks⁸, information flow through telecommunications networks⁹, and property prediction for molecular and solid materials¹⁰. From a quantum chemistry point of view, GNNs provide a unique opportunity to predict properties of solids, molecules, and proteins in a much faster way rather than by solving the computationally expensive Schrodinger equation^{11–14}.

There has been rapid progress in the development of GNN architectures for predicting material properties such as SchNet¹⁵, Crystal Graph Convolutional Neural Networks (CGCNN)¹⁶, MatERials Graph Network (MEGNet)¹⁷, Improved Crystal Graph Convolutional Neural Networks (ICGCNN)¹⁷, OrbNet¹⁸, and similar variants^{19–21}. This family of models represents a molecule or crystalline material as a graph with one node for each constituent atom and edges corresponding to interatomic bonds. A common theme is the use of elemental properties as node features and interatomic distances and/or bond valences as edge features. Through multiple layers of graph convolution updating node features based on their local chemical environment, these models can implicitly represent many-body interactions. However, many important material properties (especially electronic properties such as band gaps) are highly sensitive to structural features such as bond angles and local geometric distortions. It is possible that these models are not able to efficiently learn the importance of such many-body interactions. Explicit inclusion of angle-based information has already been shown to improve models with hand-crafted

features such as classical force-field inspired descriptors (CFID)²². Recently, there has been growing interest in the explicit incorporation of bond angles and other many-body features^{17,19,20}.

In this work, we use line graph neural networks inspired by those proposed in ref. ⁹ to develop an alternative way to include angular information to provide high accuracy models. Briefly, the line graph $L(g)$ is a graph derived from another graph g that describes the connectivity of the edges in g . While the nodes of an atomistic graph correspond to atoms and its edges correspond to bonds, the nodes of an atomistic line graph correspond to interatomic bonds and its edges correspond to bond angles. Our model alternates between graph convolution on these two graphs, propagating bond angle information through interatomic bond representations to the atom-wise representations and vice versa. We use both the bond distances and angles in the line graph to incorporate finer details of atomic structure which leads to higher model performance. Our Atomistic Line Graph Neural Network (ALIGNN) models are implemented using the deep graph library (DGL)²³ which allows efficient construction and neural message passing for different types of graphs. ALIGNN is a part of the Joint Automated Repository for Various Integrated Simulations (JARVIS) infrastructure²⁴. We train ALIGNN models for several crystalline material properties from JARVIS-density functional theory (DFT)^{24–44} and Materials project⁴⁵ (MP) datasets as well as molecular properties from QM9⁴⁶ database.

RESULTS AND DISCUSSION

Atomistic graph representation

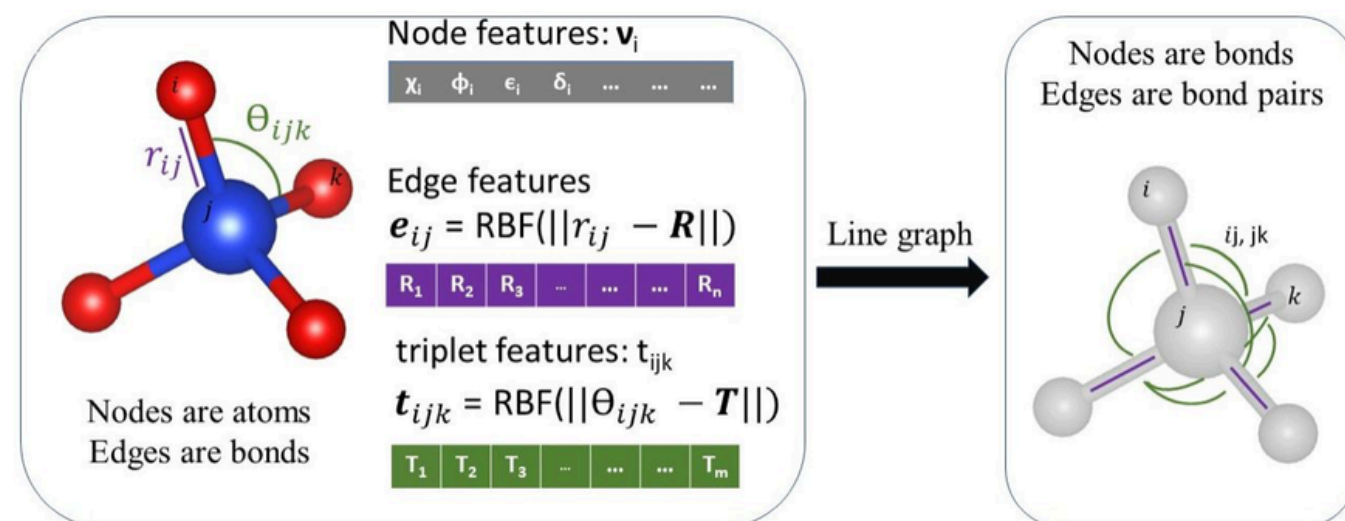
ALIGNN performs Edge-gated graph convolution⁴ message passing updates on both the atomistic bond graph (atoms are nodes, bonds are edges) and its line graph (bonds are nodes,

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DIMENET⁺⁺ (DIRECTIONAL MESSAGE PASSING NEURAL NETWORK)

→ GNN for molecular prediction
→ uses directional message passing
based on distance

distance information of atomic pairs &
angle information of atomic triplet



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Fast and Uncertainty-Aware Directional Message Passing for Non-Equilibrium Molecules

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Abstract

Many important tasks in chemistry revolve around molecules during reactions. This requires predictions far from the equilibrium, while most recent work in machine learning for molecules has been focused on equilibrium or near-equilibrium states. In this paper we aim to extend this scope in three ways. First, we propose the DimeNet⁺⁺ model, which is 8x faster and 10% more accurate than the original DimeNet on the QM9 benchmark of equilibrium molecules. Second, we validate DimeNet⁺⁺ on highly reactive molecules by developing the challenging COLL dataset, which contains distorted configurations of small molecules during collisions. Finally, we investigate ensembling and mean-variance estimation for uncertainty quantification with the goal of accelerating the exploration of the vast space of non-equilibrium structures. Our DimeNet⁺⁺ implementation as well as the COLL dataset are available online^[1].

1 Introduction

Modern machine learning models for molecular property prediction typically focus on molecules in equilibrium (e.g. QM9 [13]) or close to the equilibrium (e.g. MD17 [2], ANI-1 [15], QM7-X [7]). However, this precludes their application to the dynamics during chemical reactions, which involve transition states far away from the equilibrium. Making reliable predictions for these states requires models that are able to cover a much broader range of chemical and configurational space, i.e. including open-shell electronic structures, stretched bonds and distorted angles. In this work we aim at making progress on this problem from three directions.

First, we propose a model that is fast, accurate, and generalizes well both to different configurations and different molecules. This model predicts both the molecule's energy and the forces acting on each atom, since the latter are crucial for the molecule's dynamic behavior. To this end, we start from the recently proposed Directional Message Passing Neural Network (DimeNet) [3], which fulfills all of these properties except one: It is comparatively slow to compute. We perform a thorough model analysis to fix this and propose the DimeNet⁺⁺ model, which achieves an 8x runtime improvement while also improving predictions by 10% on average and by 20% for energies.

Second, we develop a new dataset that contains highly reactive non-equilibrium systems. The new COLL dataset contains 140 000 configurations of pairs of molecules reacting at high kinetic energies. It only consists of small molecules but covers the space of reactions much better and includes a significantly wider range of energies and forces than previous benchmarks, as shown in Fig. 4.

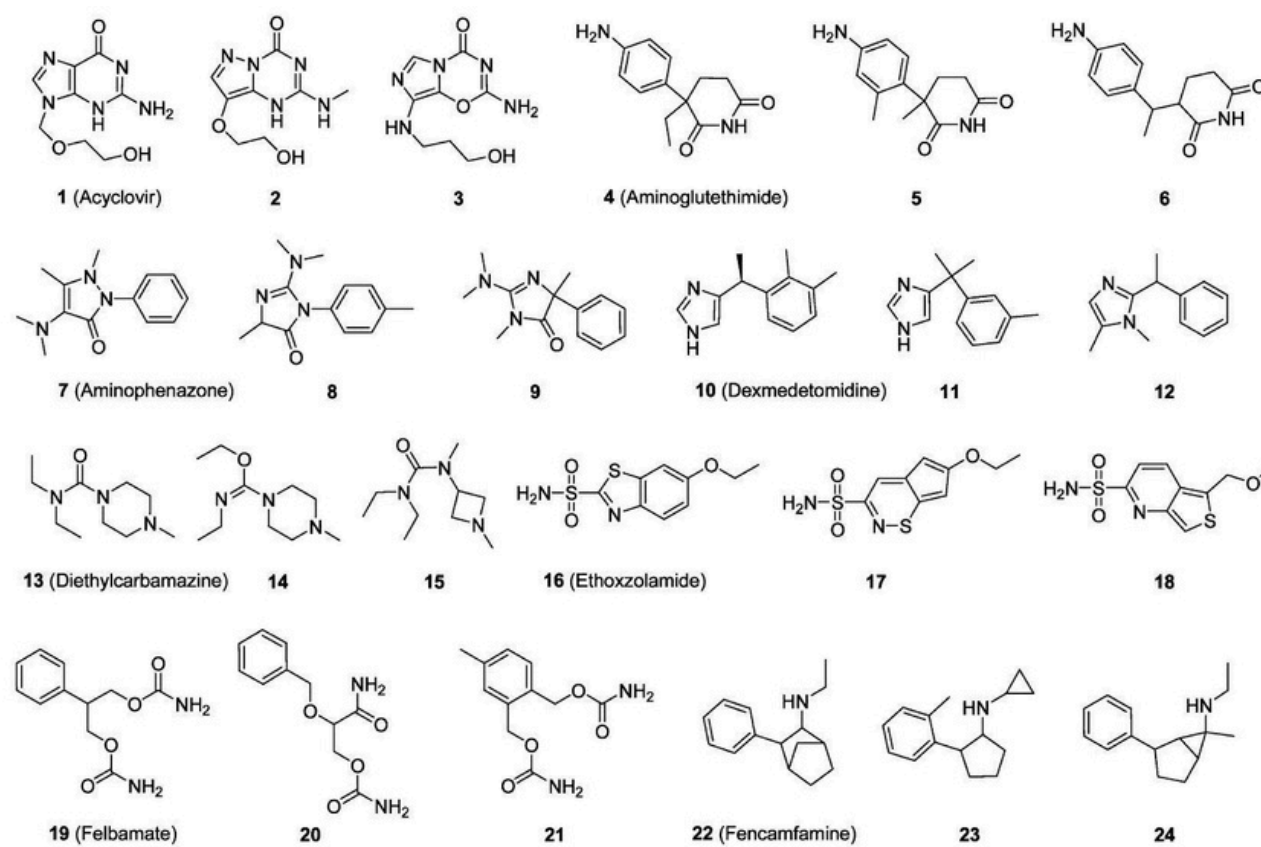
Due to the vast number of possible non-equilibrium configurations it is crucial that we are able to detect when we move out of the region covered by the training data and react appropriately (e.g. via

^[1]<https://www.daml.in.tum.de/dimenet>

Machine Learning for Molecules Workshop at NeurIPS 2020. <https://ml4molecules.github.io>

QM9

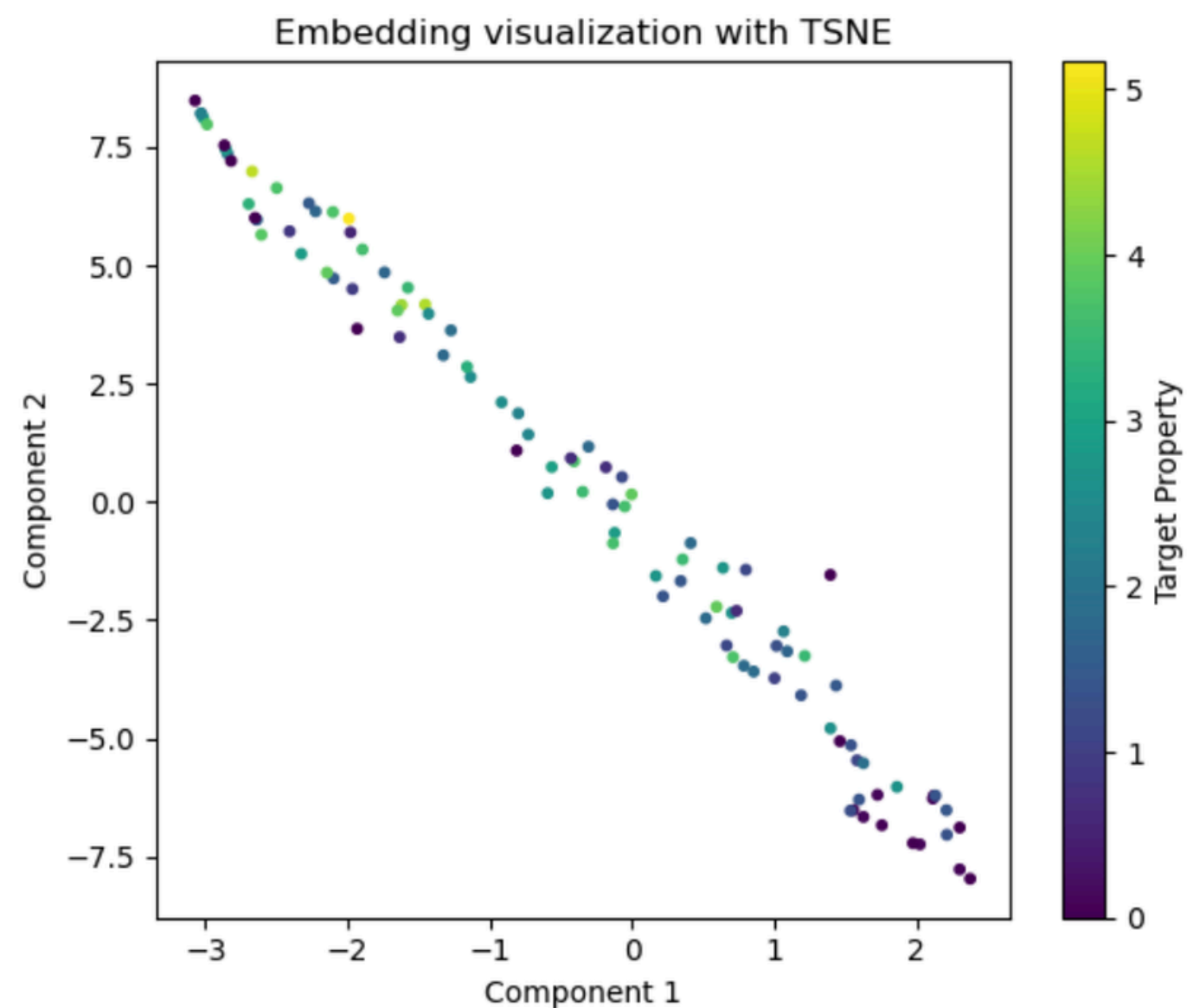
- 133886 molecules
- atoms present : H, C, N, O, F



Property	Unit	Description
μ	D	Dipole moment
α	a_0^3	Isotropic polarizability
ϵ_{HOMO}	Ha	Energy of HOMO
ϵ_{LUMO}	Ha	Energy of LUMO
ϵ_{gap}	Ha	Gap($\epsilon_{\text{LUMO}} - \epsilon_{\text{HOMO}}$)
$\langle R^2 \rangle$	a_0^2	Electronic spatial extent
zpve	Ha	Zero point vibrational energy
U_0	Ha	Internal energy at 0 K
U	Ha	Internal energy at 298.15 K
H	Ha	Enthalpy at 298.15 K
G	Ha	Free energy at 298.15 K
C_v	$\frac{\text{cal}}{\text{molK}}$	Heat capacity at 298.15 K

t-SNE

- t-SNE is for visualizing clusters or local neighborhoods.



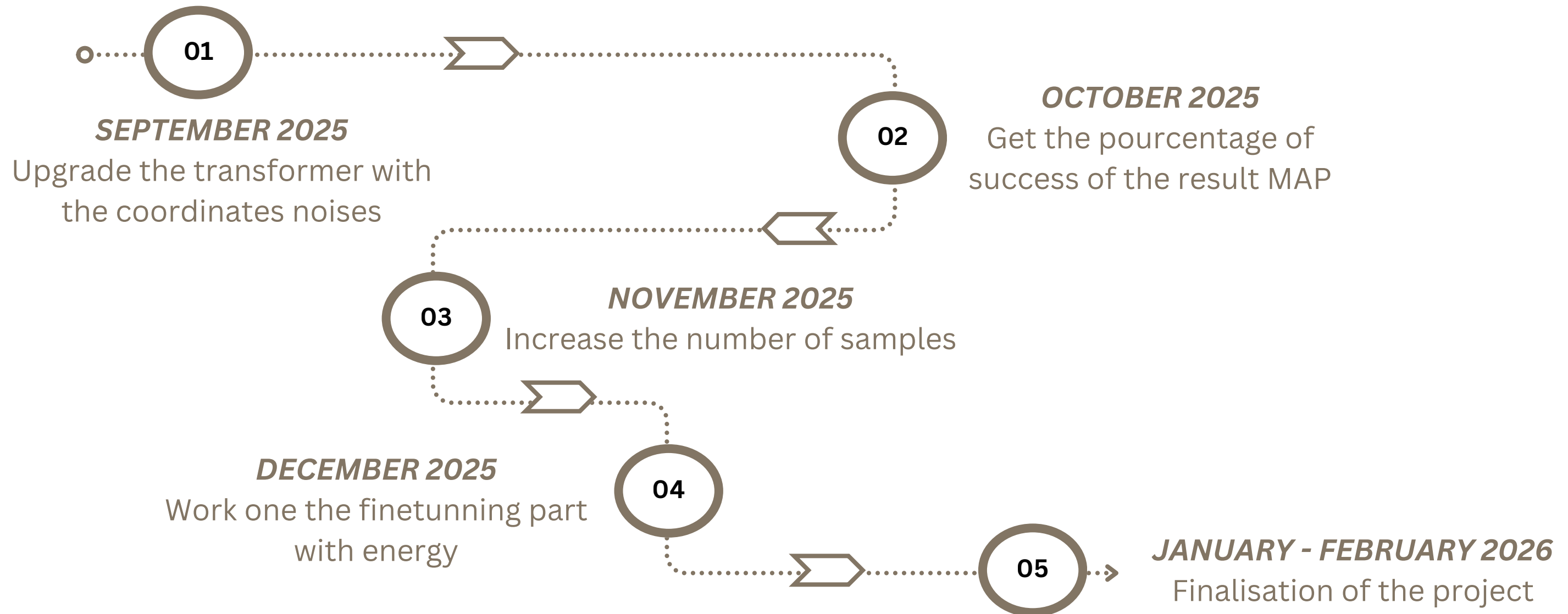
Epoch

- Decrease of the loss

```
Dimenet++ self supervised pre-training (MAP+noise)
🧠 Training DimeNet++ + Transformer...
✅ Loading pre-trained weights from pretrained_dimenetpp_map_coordnoise.pth
🔍 Extracting embeddings...
✅ pretrained_dimenetpp_map_coordnoise.pth loaded successfully.
📦 Number of molecules in dataset: 100
📊 Training downstream regressor...
Epoch 1 | Loss: 5.2567
Epoch 2 | Loss: 4.8208
Epoch 3 | Loss: 4.5127
Epoch 4 | Loss: 4.5763
Epoch 5 | Loss: 4.5380
Epoch 6 | Loss: 4.8083
Epoch 7 | Loss: 5.2240
Epoch 8 | Loss: 5.3202
Epoch 9 | Loss: 4.8618
Epoch 10 | Loss: 3.6098
```


RETROPLANNING

SEPTEMBER 2025 - FEBRUARY 2026



Thank you

BY EMILIE JIANG
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CREATIVE TECHNOLOGY
2025-2026